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SUBJECT CODE NO: - RNP-8015

FACULTY OF SCIENCE AND TECHNOLOGY

F.E (Group A & B) (NEP) Sem - I and Sem - II

Examination November / December 2025 (To be held in February-2026)

Engineering Graphics

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve any 10 question

20

- a) The symbol for first-angle projection includes:

A) A cube	B) A cone
C) A truncated cone	D) A rectangle
- b) Which one of the following is the dimensioning method?

A) Aligned system	b) Unidirectional system
c) Chain dimensioning	d) All of above
- c) When both projections of a line are parallel to the XY line, the line is:

A) Inclined to both planes	
B) Parallel to both HP and VP	
C) Perpendicular to both HP and VP	
D) Lies in both planes	
- d) Position of point in second angle

A) Observer – plane –object	B) Plane- observer-object
C) Observer-object-plane	D) Plane-object-observer
- e) Position of point in third angle

A) Above HP and Below VP	B) Below HP and Behind VP
C) Above HP and In front of VP	D) Below HP and In front of VP
- f) Position of point in fourth angle

A) Above HP and Below VP	B) Below HP and Behind VP
C) Above HP and In front of VP	D) Below HP and In front of VP
- g) If a line is inclined to both HP and VP, then:

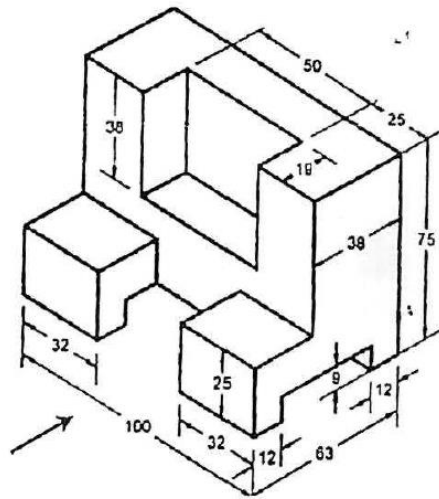
A) Its front and top views show true length
B) Its front view shows true length only
C) Its top view shows true length only
D) Both views show apparent lengths
- h) If a plane is perpendicular to the VP, its front view will be:

A) point	B) A line	C) True shape	D) Smaller than true shape
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- i) The type of line used to represent the cutting plane in drawing is.
- a) 
- b) 
- c) 
- d) 
- j) What type of views is used to provide clarity and reveal interior features of a part?
- A) Section views B) Oblique views
C) Auxiliary views D) Pictorial views
- k) Orthographic projection represents a 3D object in:
- A) Isometric view B) One single view
C) Multiple 2D views D) Perspective view
- l) The scaling factor used in isometric projection is approximately:
- A) 1 B) 0.5 C) 0.866 D) 1.732

SECTION B

- Q.2 The top view of a 75 mm. long line CD measures 50 mm. C is 50 mm in front of the V.P. and 15 mm below the H.P. D is 15 mm in front of the V.P. and is above the H.P. Draw the front view of CD and find its inclinations with the H.P. and the V.P. show its traces. **10**
- Q.3 A pentagonal plane of 30 mm side has its side on VP and 45° inclined to HP. Draw its projection when the plane is 30° inclined to VP. **10**
- Q.4 A hexagonal prism, base 30 mm side and axis 75 mm long, has a side of the base parallel to the HP and inclined at 45° to the VP. Its axis makes an angle of 60° with the HP. Draw the projection. **10**
- Q.5 A cylinder of 50 mm diameter and 60 mm height is resting on its base on HP. It is cut by a cutting plane perpendicular to VP and 45° inclined to HP and passes through mid-point of the axis of cylinder. Draw its FV, Sectional TV and True shape of the Section. **10**
- Q.6 Draw orthographic View of given object **10**
a) Front View b) Top View c) Side View



Q.7 Two views of an object are shown in the fig.. Draw its isometric view.

10

